LESSON 1.6

Math is Ours

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can describe my problem-solving process, name strategies for getting unstuck, and identify the behaviors that make our class a community of math thinkers.

WHY THIS MATTERS What is math? This lesson's answer: math is about DOING — solving problems, getting stuck, trying again, and doing it together. Think about the last time you worked with someone to solve a math problem. What did you actually do?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
representation *Spanish: representación*	A way of showing a problem — a drawing, table, equation, or model — that helps you see it.	The same 12 shown three ways: a tape diagram, a table, and 3 × 4.
strategy *Spanish: estrategia*	A plan of attack for a problem — and something you can switch when you get stuck.	"I will draw a tape diagram first, then divide" — chosen before computing.
critique *Spanish: criticar constructivamente*	To examine an idea and say what is convincing or not — about the idea, never about the person.	"That step assumes the boxes are full — how do we know?"
community agreement *Spanish: acuerdo comunitario*	A rule the whole class agrees to so that everyone can learn together.	"We take turns sharing, and we critique ideas, not people."

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

First I make sense of it: one rack holds 8 bikes, each bike has 2 wheels, so this rack has 8 × 2 = 16 wheels. The other rack has 6 times as many wheels — that is what I don't know yet.

1. I choose a representation: 6 equal groups of 16. Then I make a plan: multiply.
2. 16 × 6 = 96, so the other rack has about 96 wheels. I check my progress: 96 is 6 groups of 16, and 16 × 6 = 96, so my answer matches my plan.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

When we do math, sometimes we get stuck. What can we try?

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ comes before getting stuck because _____, but _____ only helps once _____.
- When I get stuck I can _____ .
- I can also _____ .
- The last time I worked with someone, we _____ .

Word bank representation | strategy | critique | community agreement | make sense of a problem

WATCH OUT!

A common mistake

Treating 'stuck' as a stop sign instead of a signal

LESSON 1.6

Math is Ours

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

You read a problem and have no idea where to start. According to this lesson, what does that mean?

(A) You are doing math — getting stuck is part of the process, and there are strategies to try

(B) You are not a math person

(C) The problem is broken and should be skipped

(D) You should wait silently until someone gives you the answer

STEPS TO TRY

1. What does the lesson say sometimes happens when we do math?
2. Sometimes we get stuck.
3. And then we try different strategies.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A bicycle rack holds 8 bikes, and each bike has 2 wheels. How many wheels are on the rack?

(A) 4

(B) 10

(C) 16

(D) 82

STEPS TO TRY

1. How many groups, and how many wheels in each group?
2. 8 groups of 2.
3. 8 × 2 = ?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Another bicycle rack has 6 times as many wheels as the rack with 16 wheels. About how many wheels does it have?

(A) About 22

(B) About 66

(C) About 96

(D) About 616

STEPS TO TRY

1. '6 times as many' tells you the operation.
2. 6 groups of 16.
3. 16 × 6 = ?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

Name TWO strategies from this lesson you can try when you get stuck on a problem.

SENTENCE STARTER When I get stuck I can ___ .
I can also ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Compare your two sorts. Which move belongs BEFORE you're stuck and which one only helps AFTER the plan stalls — and how would you tell a classmate which one they need right now?

SENTENCE STARTER ___ comes before getting stuck because ___, but ___ only helps once ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help